MIL-STD-2082A(AS)

26 February 1985

SUPERSEDING
(See Note)

MILITARY STANDARD

ASSEMBLY PROCEDURES, GYRO

FSC 1420

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DEPARTMENT OF DEFENSE
Washington, D.C. 20301

Assembly Procedures, Gyro

MIL-STD-2082A(AS)

1. This Military Standard is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FOREWORD

This standard has been prepared to define inspection and assembly procedures for Sidewinder Gyro Assemblies and is suitable for citation in request for proposals, invitation for bids, contracts and purchase orders, or governing specifications.

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1. SCOPE

1.1 AI Purpose. This standard describes assembly, machining, and inspection procedures for Sidewinder gyro assemblies. The gyro assembly is an integral unit of the guidance and control section, WGU-4/B and WGU-4A/B.

1.2 AI Paragraph coding. Each paragraph and table is coded as follows:

M meaning Mandatory; i.e., use instruction as stated, except for tooling.

MXM meaning Mandatory Except for Method.

AI meaning Advisory Information.

OP meaning Optional.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-C-265	Chemicals, Analytical; General Specification for.
O-E-760	Ethyl Alcohol (Ethanol); Denatured Alcohol; and Proprietary Solvent.
TT-I-1795	Ink, Marking Stencil, Opaque (Porous and Nonporous Surfaces).

MILITARY

MIL-S-22473	Sealing, Locking and Retaining Compounds; Single-Component.

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MIL-A-46050	Adhesives, Cyanoacrylate, Rapid Room-Temperature Curing, Solventless.
MIL-C-81302	Cleaning Compound, Solvent, Trichloro-trifluoroethane.
MIL-G-85105	Gyro Assembly, AIM-9M (Sidewinder).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-2081	Assembly Procedures, Gimbal.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

2104343	Lacquer, Special Optical Black.
2251012	Balance Weight, Optical Balance.
2412395	Sheet, Technical, Frequency, Specifications.
639AS516	Nut, Sunshade.
639AS534	Weight, Dynamic Balance.
639AS537	Post, Gimbal.
639AS572	Screw, Balance.
639AS577	Shim, Focusing.
639AS622	Sunshade-Damper Assembly.
639AS1467	Nut, Preload.
639AS1486	Adhesive, Epoxy-Polyamide.
639AS1487	Adhesive, Epoxy.

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639AS1574	Adhesive, Methacrylate, Solvent Solution.
639AS1611	Cement, Liquid Cellulose Nitrate.
639AS3010	Head Coil Assembly.
639AS3100	Seeker Section, Infrared, Guided Missile.
639AS3112	Mirror, Primary.
639AS3122	Aft Housing and Lens Assembly.
639AS3142	Guard, Edge.
639AS3164	Mirror, Secondary, One-Piece.
639AS3169	Gimbal Assembly.
639AS3178	Gyro Subassembly.
639AS3180	Gyro Assembly.
639AS3181	Screw, Socket Body.
639AS3182	Stud, Mirror.
639AS3184	Optical Barrel, Machined.
639AS3193	Mirror, Primary, Replica.
639AS3962	Seeker Section, Infrared, Guided Missile.
639AS6502	Gyro Assembly.
639AS6503	Gyro Subassembly.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-874	Clean Room Conditions.
AD-883	Gyro Specialist and Gyro Worker Qualifications.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AFBMA Standards	Annular Bearings Engineers Committee of the Anti-Friction Bearing Manufacturers Association (AFBMA) (Twelve Sections).

(Application for copies of AFBMA Standards may be addressed to the Anti-Friction Bearing Manufacturer's Association, 60 East 42nd Street, Rm 2825, New York, New York 10017.)

"Adhesive Bonding Techniques and Applications" by Charles V. Cagle, McGraw-Hill, 1968.

3. DEFINITIONS (Not Applicable)

4. GENERAL STATEMENT OF REQUIREMENTS

4.1 M End product. The end product gyro assembly shall comply with all listed applicable documents set forth in Section 2 of this standard and as specified herein.

4.1.1 M Production techniques. Production fabrication and assembly techniques are controlled by process sheets for all machining, inspection, and assembly phases. The instructions herein are intended to ensure acceptable quality products.

4.1.2 M Process documents. All manufacturing and inspection techniques and procedures shall be documented and made available to the particular work station involved. These procedures shall conform to the basic principles of inspection, machining, and assembly as set forth in this standard.

4.1.3 M Clean room. Clean room assembly conditions shall be in accordance with AD-874. Assembly personnel shall be qualified in accordance with AD-883.

4.2 MXM Equipment. This section lists the commercial products and equipment used during various manufacturing stages of the gyro assembly (see Drawing 639AS3180 or Drawing 639AS6502). The models or products in effect at date of issue of this standard are considered to be acceptable for use in manufacturing this assembly. However, any changes in the form of

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modifications may render the equipment unacceptable for use. Authorization to use the modified equipment may be obtained from the cognizant Government representative. Divided into two categories, the equipment and products are as follows.

4.2.1 AI Mechanical equipment.

4.2.1.1 AI Gisholt Dynamic Balancing Machine, Gisholt Model No. HS-1, Gilman Engineering and Manufacturing Company, 305 West Delavan Drive, Janesville, Wisconsin 53545. (Use Shop Accessory (SA) Tooling SA 2875049.)

4.2.1.2 AI Shrader Magnetizer, MHD4-5, Bux Shrader Company, 12080 South Western Avenue, Garden Grove, California 92641.

4.2.1.3 AI Precision Pivot Bearing (ABEC-7 or better), Miniature Precision Bearings, Incorporated, Precision Park, Keene, New Hampshire. Catalog No. S6A7B.

4.2.1.4 AI Federal Electro-Probe Electronic Test Indicator, Model 230P-3-R2, Federal Products Corporation, Providence, Rhode Island 02901.

4.2.2 AI Auxiliary items.

4.2.2.1 AI Ink, Marking Stencil, Opaque, in accordance with TT-I-1795, Type I.

4.2.2.2 AI Deltronic Tenth-Plug Gage Set, HI-Precision Grinding, Incorporated, Costa Mesa, California 92626.

4.2.2.3 AI Radio Frequency Laboratory Model 990 Magnetreater, RFL Industries, Powerville Road, Boonton, New Jersey 07005.

4.2.2.4 AI Cobehn Spray, Cobehn, Incorporated, 226 Passaic Avenue, Fairfield, New Jersey 07006. Vendor Code Ident 04882.

4.2.2.5 AI Trichlorotrifluoroethane, in accordance with MIL-C-81302, Type 1. All references to Freon within the specification are considered interchangeable with trichlorotrifluoroethane.

4.2.2.6 AI Hamilton Microliter Syringe, No. 710 with Cheney Adaptor (Hypodermic syringe with barrel graduated in microliters; No. 26 needle), Hamilton Company, P.O. Box 307, Whittier, California 90608.

4.2.2.7 AI Texwipe, Foamwipe, Foam Cubes, Texwipe Company, Hillsdale, New Jersey 07642. Vendor Code Ident 21994.

4.2.2.8 AI Primer, in accordance with MIL-S-22473 (Grade N, Form optional).

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4.2.2.9 AI Sealant, in accordance with MIL-S-22473 (Grade AA or AV).

4.2.2.10 AI Adhesive, in accordance with MIL-A-46050, Type I, Class 1.

4.2.2.11 AI Adhesive, Epoxy, in accordance with Drawing 639AS1487.

4.2.2.12 AI Adhesive Epoxy-Polyamide, in accordance with Drawing 639AS1486.

4.2.2.13 AI Tungsten Carbide Spherical Ball (0.2344 diameter) AFBMA Grade 10, Hoover Ball Division of Hoover Ball Bearing Company, Erwin Plant, Erwin, Tennessee 37650. Vendor Code Ident 29337.

4.2.2.14 AI Acetone in accordance with O-C-265, Reagent Grade.

4.2.2.15 AI Ethanol, Grade 1 in accordance with O-E-760, Class A or B.

4.2.2.16 AI Cement, in accordance with Drawing 639AS1611.

4.3 AI Solvent warning. Post the following warning notice at each work station:

WARNING AI

Prolonged breathing of solvent vapors can be dangerous. Proper ventilation is necessary in any area where they are used. Acetone and ethanol are highly flammable liquids. Fire-hazard precautions and proper ventilation are recommended in any area where they are used. Treat all solvents with caution since prolonged breathing of vapors or skin contact is injurious.

4.3.1 M Solvent purity. Use only high-purity, reagent-grade solvents that have been filtered through a 0.5 micrometer filter in all processes specified herein.

4.4 Preassembly information.

4.4.1 AI SA tooling. This section provides information on SA fixtures of recommended SA equipment used in various phases of assembly or machining. Substitutes or innovations may be used if results prove satisfactory. Familiarity with SA drawings is useful, and each operator should be acquainted with the SA tool to the extent that they know when the tool is damaged or not capable of performing the operation correctly.

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4.4.1.1 AI Naval Air Systems Command (NAVAIR) SA fixtures. SAs bearing a six- or seven-digit basic number are various forms of special tools controlled by the Naval Weapons Center (NAVWPNCEN), China Lake, California. The following is a listing of NAVWPNCEN/NAVAIR SA drawing numbers and titles:

SA 492478-3	Spanner, Tooling Gyro.
SA 492486	Test Fixture (Magnet-Mirror).
SA 492499	Balance Weight Orientation Fixture Assembly.
SA 493289	Shop Accessory (Applied to Gimbal Support Post).
SA 493292	Optical Stud Shoulder Squaring Fixture.
SA 495231	Wrench Kit, Gyro.
SA 495232	Fixture, Paint, Optical Barrel and Gimbal Assembly.
SA 496449	Primary Mirror Collimation Fixture.
SA 2875049	Dynamic Balancing Machine Fixture.

4.4.1.2 AI Vendor SA tooling. SAs bearing a seven-digit basic number are various forms of special tools designed by a vendor under Government contract and are as follows:

SA 2523825	Shim Lapping Block.

4.4.2 MXM Summary on tooling. To minimize fretting of parts, hardened tools in contact with gyro hardware are required for the gyro assembly. Keep handling tools such as jigs, fixtures, and bearing assembly devices made of magnetic materials to a minimum. This is particularly necessary when those tools will be used near the mirror-magnet. Magnetic-material tools or workbenches that come in contact with the mirror-magnet will adversely change magnet characteristics. However, in those areas where no possible chance of touching the magnet exists, hardened magnetic-material tooling may be used. Nonmagnetic tooling is used only when the magnet is present; this will avoid degrading the desired gauss field strength, particularly after the magnet has been through final magnetic stabilization.

4.4.2.1 MXM SAs and tooling inspection. Inspection of SAs and associated tooling for contamination and damage at critical surfaces is

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required at the start of every 8-hour work shift to prevent parts damage. This is the responsibility of personnel using the tooling. A 20 power (20X) minimum microscope is used for this check, where feasible, since the unaided eye cannot adequately detect minute (yet damaging) contamination and tool degradation. Successful completion of gyro assemblies is a direct function of high-level cleanliness and vigilant quality control.

4.4.2.2 M Cleanliness. Maximum cleanliness is required during all assembly periods (see 4.5.14).

4.4.3 M Component inspection.

4.4.3.1 M Contamination. Perform examination for loose foreign materials with a 10X minimum microscope. Perform necessary cleaning to remove foreign material before using hardware in assembly.

4.4.3.2 M Burrs, slivers, feather edges. Perform examination on prime datum surfaces for these defects with the aid of a 10X microscope. All remaining surfaces shall be examined for these defects with the aid of a 4X microscope. Remove loose burrs, slivers, and feather edges to preclude later contamination within the bearings.

4.4.3.3 M Nicks, dents, and bulges on functional surfaces. Perform examinations for these defects with the aid of a 10X microscope. Repair deformations such as nicks, dents, and bulges resulting in raised material on functional surfaces such that they will meet drawing requirements for surface quality and dimensional accuracy.

NOTE: AI The use of a variable power zoom microscope allows for easy finding of trouble spots at 10X magnification. Make a thorough examination of questionable areas at 20X magnification.

4.4.4 M Overhaul and repair policy. Disassembly for repair readjustment, overhaul, or replacement shall be accomplished under the same stringent cleanliness requirements as the original assembly with the added requirement that all disassembly debris shall be thoroughly and completely removed before any reassembly occurs. Aft optics lens housing may be repaired in accordance with 4.5.8.1.4.

4.5 AI Assembly data. This section provides information concerning assembly procedures.

4.5.1 MXM Adhesive application. Because successful assembly depends upon many detail-assembly procedures, the following methods of application are recommended. Strict adherence to the instructions will minimize poor

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adhesion and subsequent failure of a unit. Ensure that all mating surfaces are thoroughly dry, clean, and free from grease, oil, and other contaminants before application of any adhesive. The adhesive should wet the surface to form a satisfactory bond. Exercise discretion when applying adhesive so that contamination of functional surfaces does not occur.

CAUTION AI

Chemically blackened parts: Parts that are chemically blackened should not be subject to prolonged ultrasonic cleaning since such cleaning can be detrimental to the blackened areas.

4.5.1.1 MXM Mixing and curing adhesives. Mix and cure adhesives in accordance with manufacturer's instructions unless otherwise specified herein or on the applicable drawings. Accelerated curing temperature may be not greater than 90 degrees Celsius (°C). Use of colorant is optional.

4.5.1.2 AI Surface preparation. To check for proper surface preparation, a water-break-free test is recommended. This is a check of the surface tension of the surface to be joined. Distilled water applied with an eye dropper or equal may be used. If the surface is clean, the water will spread out and wet the surface and be free of breaks or water-break-free. If the surface is contaminated, the water will not spread or "wet" but will form droplets and have water breaks in areas where the water does not wet the surface. For a discussion of adhesive use and surface preparation, refer to the book "Adhesive Bonding Techniques and Applications," by Charles V. Cagle.

4.5.1.3 AI Sealant application. When applying sealant (see MIL-S-22473), use an appropriate-size eye dropper, graduated in microliters, so that drops of 1 microliter or 2 microliters can be applied accurately. Because the adhesive sets up quickly in a vacuum, a hypodermic-needle syringe is not recommended. The syringe depends on the vacuum principle for loading the barrel. A common sewing needle held in a pin vise may be used if it is used in a manner that ensures proper application.

4.5.1.4 AI Primer application. When applying primer (see MIL-S-22473, Grade N), a hypodermic-needle syringe is highly recommended. Use a No. 26 needle whose barrel is graduated in microliters so that drops of 1 microliter and 2 microliters can be applied accurately. After using, clean the needle and barrel with acetone (see 4.5.2). Clean and properly store both the needle and barrel when not in use.

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4.5.2 MXM Solvents. Cobehn, Freon TF, acetone, or ethanol applied with swabs as lint-free as possible are used for solvent cleaning. When using any solvent on cemented and assembled parts, exercise manual control in a manner that will not flood the spray-cleaned surfaces; such flooding will destroy uncured adhesive bonding. Since swabs are not completely lint-free, keep all swabs in clean containers until used. After using, discard the swabs in covered, disposal containers. Use only small quantities of solvent at a time. When using, ensure that the solvents are kept in small clean containers that yield clean solvents at all times. Do not use the same solvents or swabs more than once since this will result in contamination (see 4.3.1). Solvents and air used for solvent spray shall be filtered to remove particulate contamination greater than 0.5 micron.

4.5.3 MXM References. "Acetone swab," "ethanol swab," or "damp swab" mean that a dry swab or Foam Cube is dipped in the appropriate solvent in such a manner that, on application, the swab is damp but not dripping. This is accomplished by dipping a clean swab in the solvent and removing the excess by blotting on a folded Foamwipe or equal. Since bonding in many areas will be adversely affected, do not use dripping swabs during any assembly operation where adhesive damage may occur.

4.5.4 M Contamination prevention. Avoid unnecessary cleaning on mirrors, gimbal assembly, spin bearings, gimbal bearings, and the finished gyro assembly. Handle unmounted bearings in clean work stations as described in AD-874 when not in a protective container. Particular attention is drawn to the spin bearings which contain phenolic cages that are impregnated with lubricating oil to act as an oil reservoir for long storage. The bearings are grease-packed to provide a uniform film of lubricant over the working surfaces of the bearing. Cleaning with solvents will destroy both the grease-packing and the oil impregnation. Do not use a swab near enough to any bearing to cause contamination of the bearing. Relubrication of contaminated spin bearings by a qualified lubrication facility is required in the event of bearing contamination. For proper lubrication of gimbal bearings, refer to MIL-STD-2081. Do not handle bearings in any manner except with bearing handling tools or finger cots.

4.5.5 MXM Optical painted parts. Do not touch or handle optical painted surfaces unless absolutely necessary, and never with bare fingers. Fingerprints and scuff marks cause detrimental effects on gyro performance. Ensure that optical elements are kept free of grease, oil, dust, lint, etc. During assembly procedures, exercise care and remove contamination carefully when necessary.

4.5.5.1 AI Contamination. Low specular reflectance black surfaces depend in large measure on uniform unsoiled multifaceted surfaces. Contamination by body oils, other contaminants, and scuff marks due to

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excess handling raises the specular reflectance of the surface and thereby degrades the optical system.

4.5.6 M Painting. Perform all painting by spraying to obtain a smooth uniform paint coat devoid of streaks, lumps, contrasting shades, or hill-and-valley effects which create sunglint problems. Perform a spray test with each newly opened can, since paint viscosity and quality vary from one can to another. Agitate the paint in a closed container prior to use and at frequent intervals during extensive painting periods.

4.5.6.1 AI Surface preparation. Clean parts with acetone (O-C-265) or Cobehn spray to ensure that all surfaces are thoroughly dry, clean, and free from grease, oil, and other contaminates. Check for proper surface preparation in accordance with 4.5.1.2.

4.5.6.2 AI Painting. Using a pressure of approximately 7 to 13 pounds per square inch gage (psig), hold the spray gun nozzle 6 to 8 inches from the part being painted and wave nozzle back and forth. By this method the paint virtually dries before it contacts surfaces to be coated. The following criteria are used:

a. Coat is gray or shiny wet - spray gun was too close to surface.

b. Coat has velvet finish with smooth matte-like finish - spray gun position was satisfactory.

c. Coat appears like black snow - spray gun was too far away from surface.

4.5.6.2.1 OP Alternate paint process. Using a pressure of 7 to 13 psig, apply a wet coat of paint (spray gun at 2 to 6 inches) for coverage and adhesion. Air dry the wet coat of paint for 5 to 10 seconds using air from the spray gun; then back off the spray gun to a distance of 8 to 10 inches and flock spray for a rough black finish coat of paint. Immediately (within 10 to 20 seconds) after painting, carefully break bridge of paint between part and painting fixture. Oven cure as indicated in 4.5.7.1.

4.5.7 M Baking and testing paint.

4.5.7.1 M Baking. Immediately after painting parts, bake and cure the painted part at 93 ±5°C for a minimum of 6 hours unless otherwise specified herein.

NOTE: AI Whenever possible, heat-cure rather than as assemblies to avoid damage to heat-sensitive bonds and bearing lubricants.

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4.5.7.1.1 OP Alternate bake. The following parts may be bake cured at 180 ±5°C, for not less than 1 hour after painting.

a. Screw, socket body (Drawing 639AS3181).

b. Stud, mirror (Drawing 639AS3182).

c. Edge guard (Drawing 639AS3142).

4.5.7.2 M Testing. Using Cobehn solvent spray, direct a stream perpendicularly down on painted surface for 15 seconds minimum with the spray nozzle at a distance of 1 inch maximum from the painted surface. (Use 80 to 85 psig air pressure flowing through an orifice of 0.047 ±0.001-inch diameter for this test.) As used in this application, Cobehn spray is used to determine the physical adherence and durability of the paint. Perform this test on one sample piece for each 100 units (or part thereof) painted.

4.5.7.2.1 M Sample test inspection. Failure of any sample part due to poor paint adhesion or erosion shall result in lot rejection. The failed sample part shall be repaired in accordance with 3.3.8 and the entire lot shall be subjected to repainting or additional curing as necessary and retested in accordance with 3.3.7.2.

4.5.8 M Paint repair. Several parts of the gyro optical assembly require paint repair when contamination or paint damage occurs. Major paint damage due to poor adhesion, gross contamination, or surface defects shall be repaired by stripping with xylene or approved equal and repainting. Minor paint repair may be performed at piece part or assembly level by spraying or by use of an artist's brush or equal, providing that the surface quality is not affected and the part or assembly is free from undesirable paint fillets, with color optional, so final systems performance is not affected.

WARNING AI

Xylene is a highly flammable liquid. Fire hazard precautions and proper ventilation are mandatory in any area where it is used. Avoid prolonged skin contact with all solvents.

4.5.8.1 M Critical areas. The following parts require particular attention:

4.5.8.1.1 M Optical shaft. Repair allowed after assembly provided that the paint fillet between the stud and lens support shall be not greater than the allowable 0.015 maximum epoxy fillet.

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4.5.8.1.2 M Machined optical barrel. Repair allowed after assembly provided that there is no paint over spray on the primary mirror seat.

4.5.8.1.3 AI Health hazards. The damper housing assembly, which is a part of the painted sunshade-damper assembly, contains mercury-thallium, a silver colored liquid, which is a hazard to health if ingested or if the vapor is inhaled. Ensure that paint does not contaminate mounting surfaces.

WARNING AI

In case of spills or leaks of mercury-thallium, personnel are advised as follows:

a. Do not breathe the vapor; use a respirator approved for mercury.

b. Do not touch the mixture, use protective gloves.

c. Remove victims to fresh air to avoid continued inhalation; remove contaminated clothing and wash skin thoroughly. If in contact with the eyes, flush the eyes with running water for 15 minutes or more and call a physician.

d. In case of ingestion, stimulate vomiting by use of Syrup of Ipecac or touch the back of throat with finger or spoon handle, and call a physician.

e. In case of inhalation, remove victims to fresh air to avoid exposure, and call a physician.

f. Use gloves and wear a respirator approved for mercury. Pick up as much of the mercury-thallium mixture as possible using sponges and vacuum devices approved for mercury, and place in a sealed container for disposal. Dust with powdered sulfur or zinc dust. Sweep up residue and place in a sealed container for disposal.

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4.5.8.1.4 M Aft optics lens housing. Repair of minor scratches in the optical black coating on surface C of the housing is allowed using black lacquer (see Drawing 2104343), and blending the repaired area into the surrounding area.

CAUTION M

When adhesive bonds or mercury-thallium are present, do not exceed curing temperatures of 90°C.

4.5.9 M Assembly procedures. In all assembly procedures, constant vigilance and examination under 10X magnification is required to maintain contamination-free assemblies. Liberal use of vacuum-cleaning probes and fine needle-like tweezers are required for removal of contamination as it occurs. If contamination is allowed to remain for later removal, the contamination will work into the system. This may cause catastrophic failure of the assembly.

4.5.10 M Fixed axis construction. When performing optical collimation, reticle alignment, and dynamic balancing, use a precision ABEC-7 pivot ball bearing (see 4.2.1.3) at the 60-degree centerhole of the stud in conjunction with a 0.2344-inch-diameter tungsten carbide (see 4.2.2.13) spherical ball, Grade 10, in accordance with AFBMA Standards, Section 10, with the gyro rotated on its own spin bearing. The use of solid-center supports is not allowed since this contributes to degradation of the 60-degree center hole with attendant loss in accuracy.

4.5.11 MXM Optical surfaces cleaning. To clean optically finished surfaces, it is required that a nonstaining, foreign-particle-free solution be used as a solvent. Exercise extreme care when cleaning optical parts so that excessive degradation does not occur. Exercise care when using swabs. Do not scrub or use dry swabs on optical surfaces. Remove any faint stains remaining after drying. During cleaning operations, avoid degrading performance or optical properties of the surfaces.

4.5.12 M Quality assurance. Quality assurance is governed by the requirements of MIL-G-85105.

4.5.13 M Instructions. The instructions set forth in this standard and the applicable drawings apply to all contractor assembly and test procedures involving the gyro assembly. Contractor assembly and test procedures pertaining to mandatory parts of this standard are subject to review at any time by the cognizant Government representative to ensure compliance.

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4.5.14 M Assembly operations. Ensure that all assembly personnel concerned with a particular assembly operation of this standard are thoroughly familiar with that process. Assembly personnel shall be trained to be constantly alert to maintain absolute cleanliness control.

4.5.15 M Document control. Maintain a copy of this standard immediately outside clean room areas for ready reference at all times. Do not take documents inside the clean room unless they are permanently sealed in plastic containers or on vellum printed by an acceptable method.

4.5.16 MXM Machining conditions. During machining operations, filter the outside air to remove dust particles while maintaining air at a temperature between 18° to 29°C., and a relative humidity of 30 to 70 percent. Do not use coolants, and remove metal chips during the operation by means of a vacuum. Promptly clean up scrap, oil, and residue during the operation. Food and beverages shall not be allowed in the machining area.

4.5.17 MXM Ancillary requirements.

4.5.17.1 MXM Air. When warm air is referred to, it refers to air temperature of not less than 65°C but not greater than 120°C. Clean, filtered air refers to air that meets the requirements of AD-874.

4.5.17.2 MXM Solvents. Use only pure reagent grade solvents for cleaning that meet requirements of 4.3.1 for all operations.

NOTE: The procedures herein are based on the assembly of one unit at a time. Production procedures dictate more conservative use of time without violating requirements for successful assembly. Since ease of production is the primary reason for this standard, labor saving innovations may be necessary. Deviations from the basic principles are not allowed. Sequence of operations may be varied to suit production requirements provided that dimensional and functional integrity are not impaired.

5. DETAILED STATEMENT OF REQUIREMENTS

5.1 AI Optical barrel and gimbal assembly. This section describes the assembly and painting of the optical barrel and gimbal assembly.

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The use of clean, burr-free hardware throughout this and all other assembly procedures will ensure low failure rates. See 5.4 for low contamination technique.

5.1.1 AI Gimbal assembly. To assemble gimbal assembly (Item 2, Figure 1), machined optical barrel (Item 1, Figure 1), aft housing and lens assembly (Item 5, Figure 1), proceed as follows:

5.1.1.1 MXM Shipping screw and nut. Using fixture (see Drawing SA 495231), remove shipping screw, if used, and shipping nut from gimbal assembly. Do not separate the plain spin bearing from the assembly.

5.1.1.2 MXM Optical barrel. Select a machined optical barrel that provides 0.000030 to 0.000080 inch clearance between spin bearing inside diameter (ID) in the gimbal assembly and its mating outside diameter (OD) of machined lens housing. Considering that mating diameters are not perfectly round, bearing-fit clearance refers to maximum OD and minimum ID differences.

5.1.1.3 MXM Thread cleaning. Clean threads of preload nut (Item 4, Figure 1) with Cobehn or acetone. Ensure that internal threads are free of contamination when viewed under 20X magnification.

5.1.1.4 MXM Adhesive application. Apply adhesive (see Drawing 639AS1487) to internal thread of preload nut.

CAUTION M

Since adhesive will be pushed forward as female threads turn into the optical barrel assembly, only the initial engagement threads require adhesive. Do not apply adhesive to external threads of optical barrel assembly as spin bearing contamination will result.

5.1.1.5 MXM Gimbal assembly installation. Install gimbal assembly into machined optical barrel. Using fixture (see Drawing SA 495231), back off bearing retainer nut about 1/32 turn to allow radial motion for adjustment of spin bearings under preload. Using fixture (see Drawing SA 495231), install preload nut with chamfer side up onto the optical barrel. Lock with 48 to 52 inch-ounces of torque.

5.1.1.6 MXM Adhesive curing. Cure adhesive in accordance with manufacturer's instructions (see 4.5.1.1).

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5.1.1.7 MXM Torque requirements. Apply 54 to 58 inch-ounces backoff torque to preload nut. Reject the gimbal assembly if any movement occurs.

5.1.1.8 MXM Contamination examination. Remove bearing retainer nut from gimbal assembly. Examine female thread for contamination with a 20X microscope and clean out all contamination.

NOTE: AI Failure to remove all present and potential contamination from this assembly operation will result in premature failure of the spin bearing. Miniature bearings require absolute cleanliness to meet design requirements.

5.1.1.9 MXM Aft housing and lens installation. Examine male threads of aft lens housing under a 20X microscope to verify that contamination has been eliminated. Using fixture (see Drawing SA 495231), install housing and lens assembly into gimbal assembly. Lock baffle with 52 to 56 inch-ounces of torque. Apply adhesive (see Drawing 639AS1487) to two opposite slots of the housing, extending it onto the ID of the inner gimbal.

5.1.1.10 MXM Adhesive cure. Cure adhesive in accordance with manufacturer's instructions (see 4.5.1.1).

5.1.1.11 MXM Preparation for painting. Prepare optical barrel and gimbal assembly for painting by protecting the stud, gimbal, and support lens. Fixture (see Drawing SA 495232) may be used. Provide safeguards against paint spray entering spin and gimbal bearings.

5.1.1.11.1 MXM Painting and curing. Paint and cure optical barrel in accordance with 4.5.7.

5.2 AI Primary mirror and optical barrel assembly. This section describes assembly procedures for the primary mirror (Item 3, Figure 1) and optical barrel assembly (Item 1, Figure 1).

5.2.1 MXM Mirror.

5.2.1.1 MXM Magnetization. Prepare to magnetize mirror-magnet with Shrader MHD4-5 magnetizer possessing the following characteristics:

a. M Its energizing coils and cylindrical pole pieces are on the same principal axis.

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b. M It is capable of at least 50,000 ampere-turn magnetizing strength, with minimum pole-face diameter of 2.500 inches.

c. M Its magnetic path cross-sectional area, at any point, is not less than its pole-face area.

NOTE: MXM In the following, ensure that mirror-magnet is centered and is free to orient itself in the direction of the preferred magnetic field or domain. Verify that the magnet holder is nonmagnetic and that no magnetic material, other than the gyro assembly, is placed in the pole-face separation gap during magnetization.

CAUTION M

Never touch pole face with magnet.

5.2.1.2 MXM Saturation. Saturate the magnet in the magnetizer (see Drawing 2412395). In this state, ensure that the mirror-magnet generates 13.6 volts true-root-mean-square (Vtrms) minimum using fixture (see Drawing SA 492486) pickup coil while rotating at rated gyro speed (see F_1, Drawing 2412395).

NOTE: AI Magnetizing current has an effect on magnetic distortion. Satisfactory results have been obtained using a Shrader magnetizer using 20 to 25 amperes of current with poles separated 2.5 inches.

5.2.1.3 MXM Adhesive application. Apply a heavy film (approximately 0.010 thick minimum) of adhesive (see Drawing 639AS1487) to 1.124 diameter of mirror-magnet.

5.2.1.3.1 MXM Primary mirror installation. Install primary mirror into machined optical barrel. To facilitate procedures, apply a slight twist-and-push motion while installing. Remove excess adhesive with acetone or Cobehn swab so adhesive does not contaminate internal thread of optical barrel.

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CAUTION MXM

Do not physically contact optical surface of mirror during assembly.

5.2.1.4 MXM Primary mirror alignment. Align the the primary mirror to meet the requirements of Drawing 639AS3178 or Drawing 639AS6503 using fixture (see Drawing SA 496449) and gimbal post (Item 6, Figure 1). (Adjust with collimating screws of fixture, Drawing SA 496449.)

CAUTION M

Measurement is taken while the gyro optics is rotated on its own spin bearings and supported at the machined 60-degree centerhole (see 4.5.10).

5.2.1.5 MXM Adhesive curing. Cure adhesive in accordance with manufacturer's instructions (see 4.5.1.1).

5.2.1.6 MXM Reject and repair. Remove collimation fixture, check collimation to verify that no change occurs. Reject if out of specification. Repair allowed provided integrity of parts is maintained.

5.2.1.7 M Supporting barrel. Test adhesive strength by applying a 25-pound force to primary mirror (Item 3, Figure 1). Exercise care to not damage mirror surface. Reject the assembly if any relative motion between the mirror-magnet and barrel occurs. After adhesive inspection, recheck alignment in accordance with 5.2.1.4. Perform sampling for this test in accordance with MIL-STD-105, Acceptable Quality Level (AQL) of 4.0, Normal Inspection Level II.

5.2.2 MXM Gimbal post installation. Install gimbal post (Item 6, Figure 1).

5.2.2.1 OP Primer application. Apply 2 microliters of primer (see MIL-S-22473, Grade N, form optional) to internal threads of support ring. Cover threads completely by rotating part horizontally.

5.2.2.2 AI Drying. Allow ring and post to dry completely.

5.2.2.3 M Sealant application. Apply two drops (2 microliters each) of sealant (MIL-S-22473, Grade AA or Grade AV) to threads and undercut (to rear of threads) of support post; place each drop approximately 180 degrees apart. Do not contaminate forward area.

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CAUTION M

Do not apply adhesive to support ring. This will contaminate internal gimbal area which cannot be cleaned at this point of assembly.

5.2.2.4 M Support post installation. Install post permanently to optical assembly. Using fixtures (see Drawing SA 493289 and Drawing SA 492478-3), torque post to 48 to 52 inch-ounces.

5.2.2.5 MXM Excess adhesive removal. Remove excess adhesive from external surface with acetone or Cobehn swab. Curing is accomplished at completion of gyro assembly (see 5.3.18).

5.2.3 MXM Edge guard installation. Install edge guard (Item 7, Figure 1). Heat edge guard evenly to a temperature of not greater than 150°C, then install onto primary mirror. Ensure that the edge guard is firmly seated against mirror. Exercise care that no damage occurs to forward edge of edge guard and the black finish.

5.2.3.1 MXM Runout measurement. Measure runout as required by Drawing 639AS3180 or Drawing 639AS6502.

5.2.3.2 MXM Guard to mirror bonding. Bond guard to mirror using adhesive (see Drawing 639AS1487) by applying to four places equally spaced at rear of guard area.

5.2.3.3 AI Edge guard optional assembly sequence. Assembly-line handling may cause damage to the black optical finish. This assembly may be deferred to a later sequence step such as the final part to be assembled, if necessary, to protect the finish against handling damage.

5.2.4 M Magnet stabilization. Install assembly on magnetreater, stabilize the mirror-magnet to 12 ±0.25 Vtrms as generated using fixture (see Drawing SA 492486) pickup coil while gyro is rotating at rated gyro spin speed. Stabilize all units at a voltage at least 10 percent below the saturation voltage. Verify that the generated voltage sine wave has less than 3 percent distortion.

5.2.4.1 MXM Stabilization. Accomplish stabilization by inserting the mirror-magnet into the processing coil of a Radio Frequency Laboratory Model 990 Magnetreater, or equal (see 4.2.2.3).

5.2.4.2 MXM Remagnetization. If overstabilization occurs, remagnetize to saturation and repeat 5.2.4.

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5.2.4.3 AI Installation of gyro into head coil. Install gyro into a head coil and rotate at a spin frequency of F_1 (see Drawing 2412395). Monitor reference voltage at pins J601-3 and J601-4 on an oscilloscope. The sine wave is to have a uniform slope near the zero crossover point. If the slope is not uniform, demagnetize the magnet and repeat steps 5.2.1.2 and 5.2.4 through 5.2.4.3.

5.2.4.4 AI Installation verification. Verify that none of the rotating parts touch any of the stationary parts.

5.2.5 MXM Total indicated runout (TIR). Check runout of secondary mirror seating face of the hub and mirror assembly. If the TIR is greater than 100 microinches, use fixture (see Drawing SA 493292) to perform a collimation cut on the seating face (see 4.5.16). Perform cut in an isolated, clean, hooded area so that the conditions of AD-874 are not violated in the clean room. Fixture (see Drawing SA 493292) contains an exhaust line to remove metal chips allowing the isolated clean hood to also meet conditions of AD-874 with the single exception that metal removed is allowed in this specific case only. Ensure that the cut surface is concave to obtain proper setting of the mirror assembly. Hold metal removal to 0.001 maximum per machining pass.

5.3 MXM Final gyro assembly (Figure 2).

5.3.1 MXM Focusing shim installation. Install a nominal focusing shim (Item 2, Figure 2) to optical assembly.

5.3.2 MXM Secondary mirror installation. Install a secondary mirror (Item 3, Figure 2) to optical assembly. Ensure that scribe mark on mirror is 68 ±5 degrees counterclockwise (CCW) to the north magnetic pole of mirror-magnet when observer is looking into primary mirror.

5.3.3 MXM Sunshade and damper assembly installation. Install sunshade and damper assembly (Item 4, Figure 2) (see 4.5.8.1.3) to optical assembly. Position the sunshade and damper assembly such that the seal plug in the outermost channel is 90 ±20 degrees CCW to scribe mark on the secondary mirror when looking into the primary mirror.

5.3.4 MXM Sunshade nut installation. Install sunshade nut (Item 5, Figure 2) and torque to 30 to 32 inch-ounces.

5.3.5 OP Dynamic balancing. Using Gisholt HS-1 balancing machine, perform preliminary dynamic balance of gyro optical assembly to a level that will permit focusing (see 4.2.1.1 and 5.3.13).

5.3.6 MXM Focus adjustment. Adjust focus to meet the requirements of MIL-G-85105 by using fixture (see Drawing SA 493292) to cut the face of the stub (-2 configuration) or adjust through the use of a single shim

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(-1 configuration) (the exposed surface shall be black) whose thickness provides optimum focus.

a. AI Check shim parallelism. Drawing 639AS577 requires parallelism within 50 microinches. Fixture (see Drawing SA 2523825) may be used to lap focus shims to 25 microinches parallelism. Blacken the shim edge (circumference).

b. AI Recheck the cut taken in 5.2.5 for accuracy. Ensure that the seat is within 100 microinches TIR.

c. AI If the post is overmachined (-2 configuration), cement a shim of excessive thickness to the face of the mirror post using adhesive (see Drawing 639AS1487). Then, machine the shim to meet the optimum focus requirements of MIL-G-85105.

5.3.7 M Final assembly. When optimum focus has been achieved, complete final assembly of focus shim, secondary mirror, sunshade, and sunshade nut (see Figure 2).

5.3.7.1 MXM Installation. Install parts in the following order:

a. M Focus shim, if used. Verify that the black finish has not been damaged.

b. M Secondary mirror. Scribe mark on mirror shall be 68 ±5 degrees CCW to the north magnetic pole of mirror-magnet.

c. M Sunshade and damper assembly to optical assembly. Position the sunshade and damper assembly until the seal plug in the outermost channel is 90 ±20 degrees CCW to the scribe mark on the secondary mirror when looking into the primary mirror.

d. M Sunshade nut. Torque sunshade nut in accordance with 5.3.4.

5.3.8 MXM Mirror positioning. Verify that the low spot (scribe mark) of mirror is 68 ±5 degrees CCW to the north magnetic pole of mirror-magnet.

5.3.9 MXM Adhesive application. Apply adhesive (see Drawing 639AS1487) to two slots of sunshade nut. Remove excess adhesive from front face of nut. Cure adhesive in accordance with 4.5.1.1. Leave room in slots for applying backoff torque.

5.3.10 MXM Adjustment. Using gravity balance screw (Item 6, Figure 2), adjust fore and aft control.

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CAUTION M

When installing balance screw, do not violate 0.088-inch minimum clearance from stud forward face to balance screw forward face (see Figure 2). This clearance is required to permit use of a pivot ball bearing for dynamic balancing phases. Do not cement screw at this time; adjustments may be required later (see 5.3.17).

5.3.11 MXM Fore and aft balance adjustment. Adjust the fore and aft balance of the gyro optical assembly to meet the requirements of MIL-G-85105. The test fixture used shall incorporate a simulated missile, including a head coil assembly, requisite electronics circuitry, and a rotating device. Perform the test with the gyro optical assembly mounted in the head coil assembly on the rotating device, and with the gyro rotating at regulated speed with the spin axis of the gyro and support post in a horizontal plane. Monitor the two current inputs to the servo amplifiers while the head coil assembly is rotated about the gyro spin axis at a rate of 4.0 ±0.2 degrees per second. Compute the mass unbalance drift rate from the current input to the servo amplifiers. If the computed drift rate is greater than 0.025 degree per second, readjust the gravity balance in accordance with 5.3.10. Repeat the testing and readjustment alternately, as many times as necessary, to obtain a drift rate not greater than 0.025 degree per second in each input to the servo amplifier.

5.3.12 MXM Balance weight cleaning. Clean dynamic balance weights (Item 7, Figure 2) with acetone or Cobehn.

5.3.12.1 MXM Coating. Coat the outside spherical diameter of the mirror magnet with ink, marking stencil, opaque, in accordance with TT-I-1795, Type I, color black or blue starting at the north pole within 5 degrees proceeding clockwise, when looking into mirror face. Coat the spherical diameter as required to activate the photocell of the balance equipment (see 5.3.13.1).

5.3.13 M Dynamic balance adjustment. Perform dynamic balance adjustment as follows:

5.3.13.1 M Procedure. Set up gyro optical assembly in dynamic balancing machine, using fixtures (see Drawing SA 2875049 and Drawing SA 492499), and a Gisholt HS-1 balancing machine as adapted for gyro optical assembly. While balancing, use a gyro speed of 5,000 ±50 revolutions per minute (rpm) for consistency during checking at various activities.

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5.3.13.1.1 MXM Calibration. Periodically calibrate the dynamic balancing machine as follows:

NOTE: MXM Calibration as outlined in 5.3.13.1.1.2.1 is performed once a month. Calibration of the 150 and 300 micro-ounce-inch points is performed once a week at a time agreeable to the Government representative or at any time the Government representative deems necessary to ensure quality balancing. Table I contains a list of required equipment. Also refer to Tables II and III. Tables IV and V show estimated values for production. This test is performed with a different standard production gyro for each calibration period. The use of master gyros for calibration is acceptable as a reference only.

TABLE I. MXM Required equipment for calibration.

	Description
1.	Standard production gyro (master gyro optional for reference).
2.	Calibration weights as indicated in Table II.
3. 1/	Adhesive, MIL-A-46050, Type I, Class 1 or an equal fast-drying cement that permits quick application in minute amounts. Cement must be able to be removed easily upon completion of test.
4.	Dynamic balancing machine with an amount meter calibrated as indicated in 5.3.13.1.1.1.
5.	Fixture, Drawing SA 492499, Weight Orientation Fixture.

1/ MIL-A-46050 cement is quickly removed by using a small electrical soldering iron or electrical soldering pencil held briefly against weights applied to the gyro, since temperatures of approximately 170°C will rapidly weaken the bond.

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TABLE II. MXM Master weights for balancing planes.

Weight in micro-ounces at mean pitch radius of 0.606 inch	Primary mirror balance plane		Weight in micro-ounces at mean pitch radius of 0.4604 inch	Damper housing balance plane	
	Length of weight 1/	Unbalance in micro-ounce inches		Length of weight 1/	Unbalance in micro-ounce inches
248	0.080 2/	150	326	0.0106 2/	150
495	0.160 2/	300	652	0.211 2/	300
990	0.321 2/	600	1303	0.130 2/	600
1980	0.198 3/	1200	2606	0.261 3/	1200
2970	0.297 3/	1800	3910	0.391 3/	1800

1/ Lengths given are reference only. Weights in columns 1 and 4 are required for calibration.

2/ Based on aluminum wire.

3/ Based on copper wire.

TABLE III. AI Data on which Tables IV and V are based.

(1) Material

(a) Copper bus bar - calibration standard weight (CSW_{Cu}):
Weight in micro-ounces per inch = 9,995.71
Mean diameter in inches = 0.051

(b) Aluminum - calibration standard weight (CSW_{Al}):
Weight in micro-ounces per inch = 3,085.60
Mean diameter in inches = 0.051

(c) Lead pig - calibration standard weight (CSW_{Pb}):
Weight in micro-ounces per inch = 13,889.57
Mean diameter in inches = 0.051

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bracket (MXM 001-1-0141) shall be removed from the fire extinguisher

TABLE III. AI Data on which Tables IV through V are based (continued).

(2) Primary mirror balance plane mean pitch radius = 0.606 inch is based on:

	1/263	ID of balance shoulder
minus	0.051	balance weight diameter
	1.212	mean pitch diameter

0.606 = mean pitch radius

(3) Damper housing balance plane mean pitch radius = 0.4604 inch is based on:

	0.9718	ID of balance shoulder (0.662 +(0.971 - 0.662)cos 4 degrees)
minus	0.051	balance weight diameter
	0.9208	mean pitch diameter

0.4604 = mean pitch radius

(4) Let X = weight required to correct for balance tolerance. Then X times $\text{radius}_{\text{mean}}$ = balance tolerance in micro-ounce inches.

Therefore:

$\frac{X}{CSW}$ = length of weight to correct for tolerance.

(5) Let Y = weight required to correct for unbalance. Then Y times $\text{radius}_{\text{mean}}$ = balance correction in micro-ounce inches.

Therefore:

$\frac{Y}{CSW}$ = length of weight to correct for unbalance.

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TABLE IV. AI Primary mirror balancing plane (production table).

(Recommended to keep correction weights within reasonable length.)
(Rounded off to nearest thousandth.)

Unbalance in micro-ounce inches	Dial mark	Weight length in inches	Unbalance in micro-ounce inches	Dial mark	Weight length in inches
Aluminum Range					
50	1	0.027	450	9	0.241
100	2	0.054	500	10	0.267
150	3	0.080	550	11	0.294
200	4	0.107	600	12	0.321
250	5	0.134	650	13	0.348
300	6	0.160	700	14	0.374
350	7	0.187	750	15	0.401
400	8	0.214	...	...	...
Copper Range					
800	16	0.132	1700	34	0.281
850	17	0.140	1750	35	0.289
900	18	0.149	1800	36	0.297
950	19	0.157	1850	37	0.305
1000	20	0.165	1900	38	0.314
1050	21	0.173	1950	39	0.322
1100	22	0.182	2000	40	0.330
1150	23	0.190	2050	41	0.338
1200	24	0.198	2100	42	0.347
1250	25	0.206	2150	43	0.355
1300	26	0.215	2200	44	0.360
1350	27	0.223	2250	45	0.371
1400	28	0.231	2300	46	0.380
1450	29	0.239	2350	47	0.388
1500	30	0.248	2400	48	0.396
1550	31	0.256	2450	49	0.405
1600	32	0.264	2500	50	0.413
1650	33	0.272			

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TABLE IV. AI Primary mirror balancing plane (production table) (continued).

(Recommended to keep correction weights within reasonable length.)
(Rounded off to nearest thousandth.)

Unbalance in micro-ounce inches	Dial mark	Weight length in inches	Unbalance in micro-ounce inches	Dial mark	Weight length in inches
Lead Range					
500	1 1/	0.059	3000	6 1/	0.356
1000	2 1/	0.119	3500	7 1/	0.416
1500	3 1/	0.178	4000	8 1/	0.475
2000	4 1/	0.238	4500	9 1/	0.535
2500	5 1/	0.297	5000	10 1/	0.594

1/ Expanded scale. Each mark = 500 micro-ounce inches.

TABLE V. AI Damper housing balancing plane (production table).

(Recommended to keep correction weights within reasonable length.)
(Rounded off to nearest thousandth.)

Unbalance in micro-ounce inches	Dial mark	Weight length in inches	Unbalance in micro-ounce inches	Dial mark	Weight length in inches
Aluminum Range					
50	1	0.035	300	6	0.211
100	2	0.070	350	7	0.246
150	3	0.106	400	8	0.282
200	4	0.141	450	9	0.317
250	5	0.176	500	10	0.352

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TABLE V. AI Damper housing balancing plane (production table).

(Recommended to keep correction weights within reasonable length.)
(Rounded off to nearest thousandth.)

Unbalance in micro-ounce inches	Dial mark	Weight length in inches	Unbalance in micro-ounce inches	Dial mark	Weight length in inches
Copper Range					
550	11	0.120	1050	21	0.228
600	12	0.130	1100	22	0.239
650	13	0.141	1150	23	0.250
700	14	0.152	1200	24	0.261
750	15	0.163	1250	25	0.272
800	16	0.174	1300	26	0.283
850	17	0.185	1350	27	0.293
900	18	0.196	1400	28	0.304
950	19	0.206	1450	29	0.315
1000	20	0.217	1500	30	0.326
Lead Range					
1550	31	0.242	2050	41	0.321
1600	32	0.250	2100	42	0.328
1650	33	0.258	2150	43	0.336
1700	34	0.266	2200	44	0.344
1750	35	0.274	2250	45	0.352
1800	36	0.282	2300	46	0.360
1850	37	0.289	2350	47	0.368
1900	38	0.297	2400	48	0.375
1950	39	0.305	2450	49	0.383
2000	40	0.313	2500	50	0.391
500	1 1/	0.078	3000	6 1/	0.469
1000	2 1/	0.156	3500	7 1/	0.547
1500	3 1/	0.235	4000	8 1/	0.626
2000	4 1/	0.313	4500	9 1/	0.704
2500	5 1/	0.391	5000	10 1/	0.782

1/ Expanded scale. Each mark = 500 micro-ounce inches.

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5.3.13.1.1.1 MXM Requirements for calibration.

a. MXM Gyro speed: 5,000 ±50 rpm (maintain speed constant within ±10 rpm).

b. MXM Amount meter: Calibrated in units such that unbalance down to 25 micro-ounce inches can be read easily.

c. MXM Master weights: A complete set of accurately measured and weighed master weights.

d. MXM Equipment listed in Table I.

5.3.13.1.1.2 MXM Calibration test. Conduct calibration test as follows:

5.3.13.1.1.2.1 MXM Use of standard. Use a gyro that has been balanced within 50 micro-ounce inches in each plane. Record the angle and amount for each plane of correction.

5.3.13.1.1.2.2 MXM Equipment accuracy. Demonstrate the accuracy of the equipment by using the series of master weights shown in Table II as follows: Select three positions on the magnet and three positions on the sunshade until the master weights can be placed 120 degrees apart and not interfere with the existing balance weights. The master weights of Table II shall be placed in succession at each angular position so that a series of measurements at each angle in each plane will be made from an unbalance of 150 to 300 micro-ounce inches; i.e., the weights shall be placed at one position only and in one plane at a time. The weights shall be fastened with a minimum of cement so the calibration of the standard weights cannot be altered. The following requirements shall be met:

a. MXM Accuracy. The amount meter to be accurate within 25 micro-ounce inches of the vector sum of the amount of known unbalance being applied plus the original unbalance when the weight is applied at the angle shown for the original unbalance.

NOTE: MXM Assume the original unbalance reading is incorrect. This will be exposed during use of the first test weight since the machine will be shown to be nonlinear. Use of the other test weights is to demonstrate linearity over the whole dial range within the range of average dynamic unbalances encountered in the gyro.

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b. MXM When testing each weight in a given plane (magnet or sunshade), a check will simultaneously be made of the opposite plane to demonstrate that not greater than 10 percent crosstalk exists between planes. Select a weight which will produce an unbalance indication of not less than 50 percent of full scale on the meter.

c. MXM The angle indicator reading to be accurate within ±10 degrees of the known angle at which the known weight was applied

d. MXM Failure to meet the preceding test specifications requires that the balancing machine be readjusted and then retested to the satisfaction of the Government representative

5.3.13.2 M Balancing. Hold gimbal in one fixed horizontal position during balancing. This requires a two-point support at stud centerhole and at gyro support post in accordance with 4.5.10.

CAUTION AI

Worn out pivot balls (0.2344 diameter) can cause erratic results during calibration, collimation, and dynamic balancing operations. Use several balls during monthly calibration to expose potential problems.

5.3.13.3 M Two-plane correction. Planes of correction are forward face of sunshade and aft side of mirror-magnet (see Drawing 639AS3180 or Drawing 639AS6503 for plane location). Cement weights (Item 7, Figure 2) in place with adhesive (see MIL-A-46050) or cement (see Drawing 639AS1611) after proper weight and angular location are determined. Push the weights firmly into place so as to contact both surfaces of the magnet (see 5.3.14). Tolerance in each plane is 150 micro-ounce inches maximum. Use fixture (see Drawing SA 492499) for proper location of correction weights. After drying, verify that the balance is within specification.

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CAUTION M

When moving weights or cleaning excess adhesive, thin filaments of adhesive may be produced which may contaminate assembly if care is not exercised.

CAUTION M

Any physical changes to gyro-optic spinning mass (such as phasing, focusing, and large optical corrections) require that gyro be rechecked through dynamic balancing to ensure that it still lies within specification.

5.3.13.3.1 MXM Adhesive and paint application. After completion of dynamic balance, apply adhesive, methacrylate (see Drawing 639AS1574) for a permanent bond and paint weights in forward plane with art brush, or equal, using black optical lacquer in accordance with Drawing 2104343 or black ink in accordance with TT-I-1795, Type 1. Use extreme care to prevent damage to remainder of assembly.

5.3.14 M Verification. Verify that no rotating part touches any of the stationary parts.

5.3.15 M Optical scan diameter. Perform scan diameter measurement (optical balance). Verify that the gyro assembly (see Drawing 639AS3180 or Drawing 639AS6502), as installed in the seeker section (see Drawing 639AS3100), meets the scan diameter requirement of MIL-G-85105. If out of specification, adjust as follows.

5.3.15.1 MXM Cleaning. Clean optical balance weights (Item 8, Figure 2) with acetone or Cobehn.

5.3.15.1.1 M Gyro assembly adjustment. Using the proper amount of weight for optical balance, adjust gyro assembly by installing weights in sunshade so the requirements of 5.3.15 are met. Use cement technique as specified in 5.3.13.3 and paint weights as specified in 5.3.13.3.1. Observe CAUTION note of 5.3.13.3, except that after optical balance has been completed, the final dynamic specification for each plane is as follows:

Magnet plane - 155 micro-ounce inches maximum.

Sunshade plane - 200 micro-ounce inches maximum.

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5.3.16 AI Excess cement removal. Remove excess cement with acetone (see CAUTION note of 5.3.13.3).

5.3.17 MXM Gravity balance screw cementing. Repeat 5.3.10 as required, and carefully cement gravity balance screw with adhesive described on Drawing 639AS1574. Cementing sequence for this screw is optional. If this screw is removed for any reason by either dissolving cement or brute force, damage can occur which may reject the gyro. Alternate method of cementing screw is to apply adhesive to form a fillet on top of the screw.

CAUTION M

When installing balance screw, do not violate the 0.088-inch minimum clearance from stud forward face to balance screw forward face (see Figure 2). This clearance is required to permit use of a pivot ball bearing for dynamic balancing phase.

5.3.18 M Gyro assembly curing. Cure completed gyro assembly for minimum of 24 hours at room temperature.

5.3.19 M Torque. Apply CCW backoff torque to the following parts in such a sequence that damage does not occur to the assembly:

a. M Support post (Item 6, Figure 1). Apply 54 to 58 inch-ounces.

b. M Sunshade nut (Item 5, Figure 2). Apply 34 to 38 inch-ounces.

c. M Gravity balance screw (Item 6, Figure 2). Apply 6 to 10 inch-ounces.

5.3.19.1 M Rejection. Reject the assembly if any CCW motion occurs.

5.3.20 M Inspection. Inspect for paint damage and examine all visible areas of the gyro assembly under 10X minimum magnification for contamination in excess of the following acceptance criteria:

a. Nonmetallic contamination - not greater than 0.003 inch.

b. Metallic contamination - not greater than 0.001 inch.

5.3.21 M Storage. Store completed gyro assembly in closed dust-free container until assembly is installed in the seeker head of the seeker section (see Drawing 639AS3100 or Drawing 639AS3962) where it is then permanently sealed against the outside atmosphere.

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5.4 AI Alternate low contamination assembly technique for optical barrel and gimbal assembly.

5.4.1 AI Contamination. This technique may be used when there is a persistent spin bearing contamination problem at the assembly level. This low contamination assembly technique is based on the fact that contamination is falling into the bearing during assembly operations, such as screwing threads into position, thereby generating debris. This type of contamination can be prevented from entering the spin bearing by working in a vertical position with the optical barrel on top and the gimbal below.

a. AI The preload nut, when installed with adhesive inside, contributes little if any contamination and retains within the adhesive any generated contamination.

b. AI The housing and lens assembly (aft lens housing) cannot discharge generated contamination into the bearing once it is seated.

c. AI Special fixturing is recommended if the vertical approach is used to keep down cost.

5.5 AI Supersession note. This standard supersedes the following:

a. MIL-STD-2082(AS) dated 17 October 1980.

b. Notice 1, dated 17 March 1982.

c. ECP M7531R1C1 (NOR M7531.2A) dated 5 November 1981.

d. ECP F1045R1C1 (NOR F1045.1A) dated 14 January 1983.

e. ECP F1088R1 (NOR F1088.1A) dated 23 February 1983.

f. ECP 11254 (NOR 11254.1).

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(Project 1420-N273)

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1	639AS3184	OPTICAL BARREL, MACHINED
2	639AS3169	GIMBAL ASSEMBLY
3 1/	639AS3112	PRIMARY MIRROR
4	639AS1467	PRELOAD NUT
5	639AS3122	AFT HSG AND LENS ASSY
6	639AS537	GIMBAL POST
7	639AS3142	EDGE GUARD 2/

1/ ALTERNATE: 639AS3193

2/ THE EDGE GUARD (ITEM 7) MAY BE INSTALLED AT THIS LEVEL OF ASSEMBLY OR AT THE FINAL ASSEMBLY LEVEL

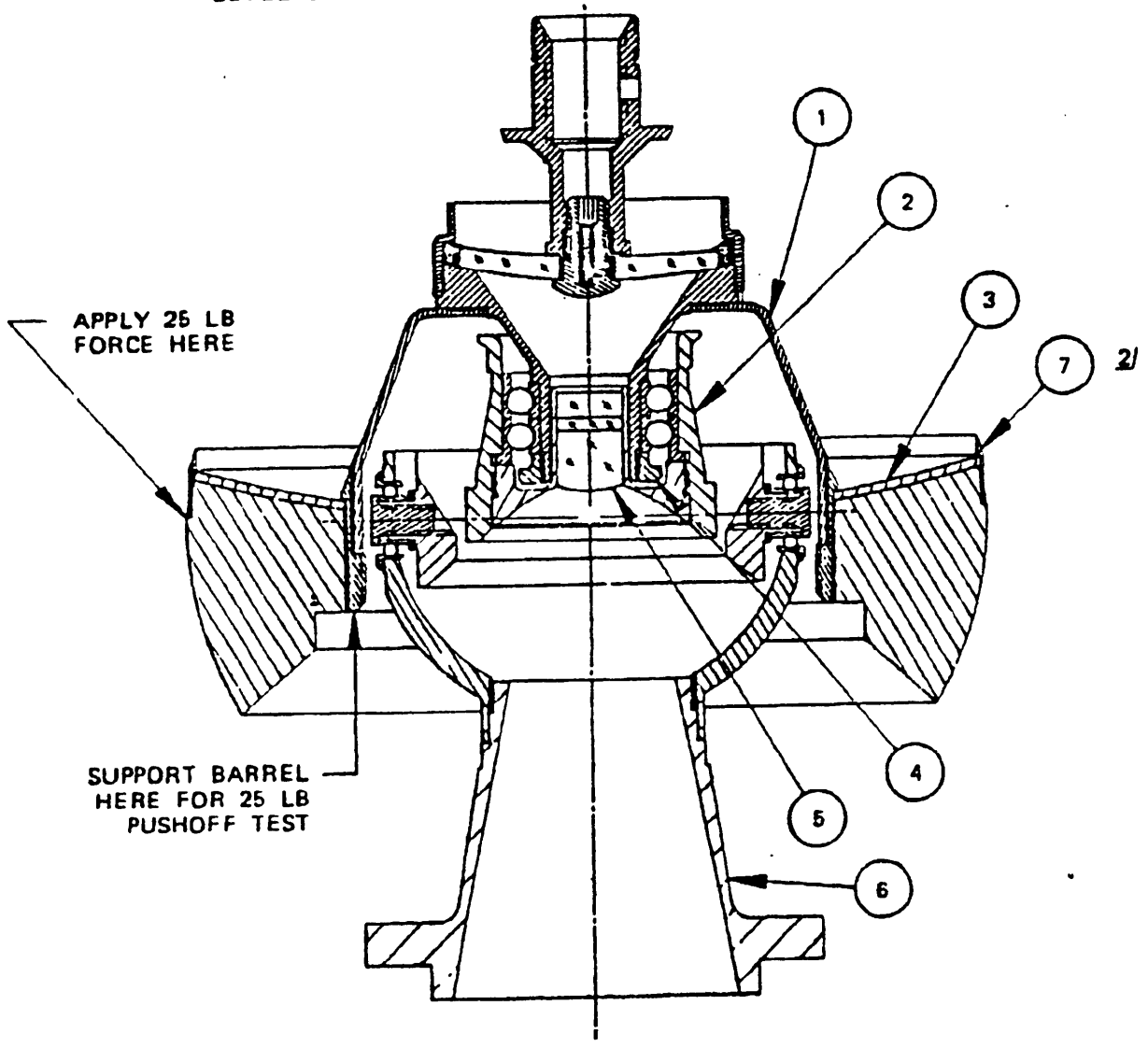


FIGURE 1. Gyro subassembly (Drawing 639AS3178 or Drawing 639AS6503.

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ITEM	PART NUMBER	NOMENCLATURE
1	639AS3178 OR 639AS6503	GYRO SUBASSEMBLY
2	639AS577	FOCUS SHIM
3	639AS3164	SECONDARY MIRROR
4	639AS622	SUNSHADE–DAMPER ASSY
5	639AS516	SUNSHADE NUT
6	639AS572	GRAVITY BALANCE SCREW
7	639AS634	DYNAMIC BALANCE WEIGHT
8	2251012	OPTICAL BALANCE WEIGHT

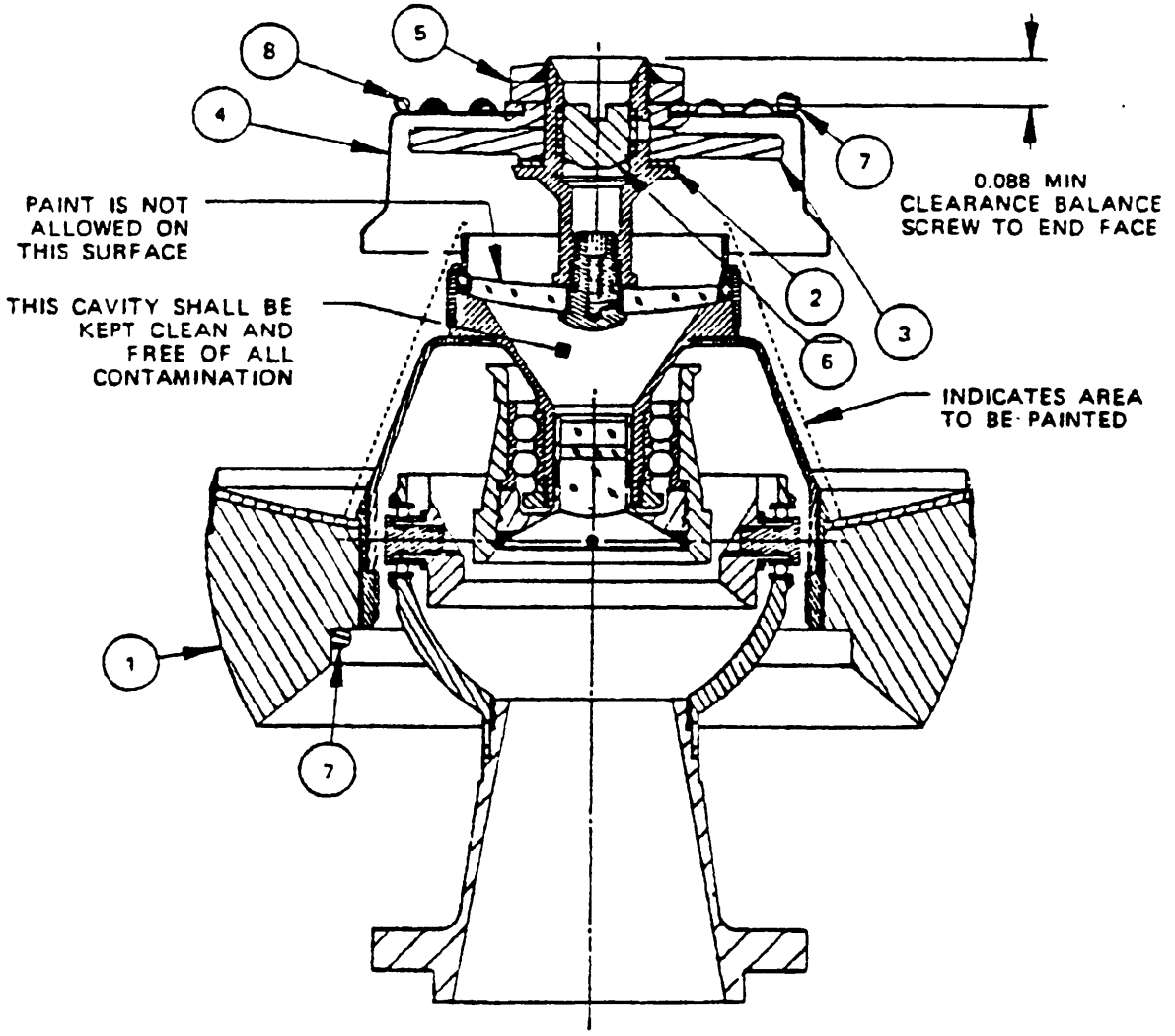


FIGURE 2. Gyro assembly (Drawing 639AS3180 or Drawing 639AS6502).

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